3D Printing and Modeling

1st lab: Introduction to our laboratory and basics of Fusion 360

0. Resources

1. Fusion 360 Install: Install Fusion 360 free with student account
2. Prusa Slicer Install
3. Fusion 360 User Interface Tutorial (~ 5 min)
4. How to create a rectangle in Fusion 360

0.5. Lab 414

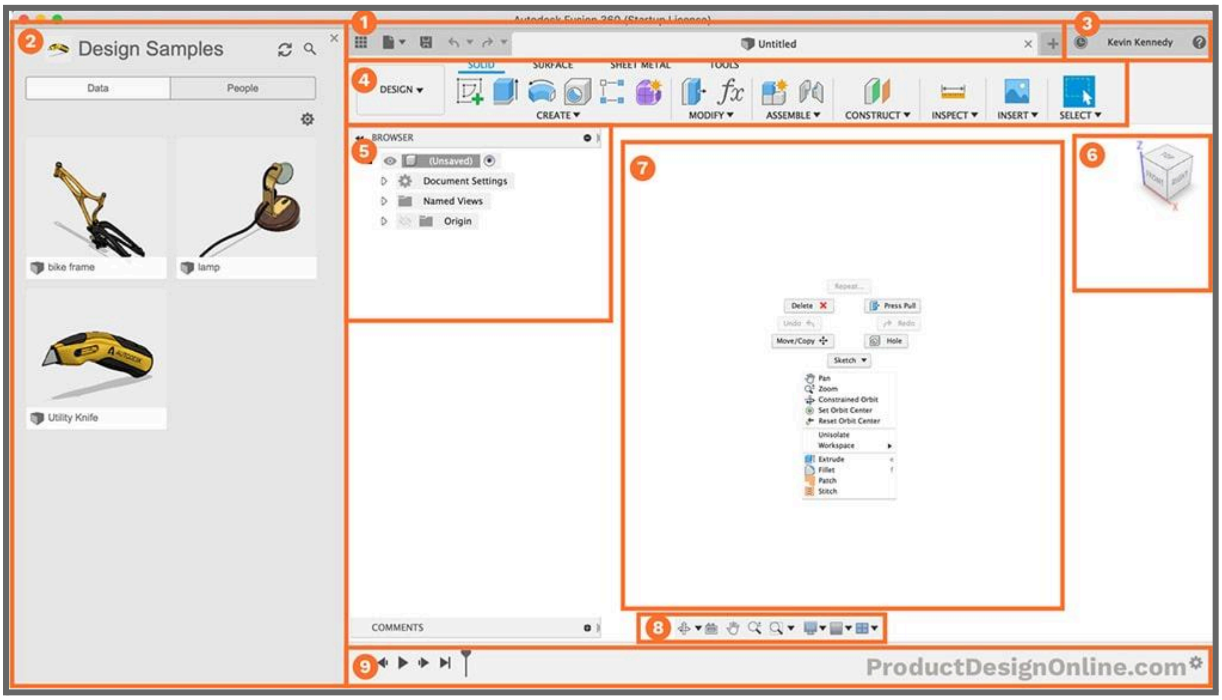
1. Short introduction on what our lab contains:
    a. Our 3D Printers
    b. The filament cabinet
    c. The wall of tools
    d. The projects area

> The lab is an open space where you can come and work on any projects at any time during the semester. However, please keep the space clean and be respectful of others.

1. Fusion 360

Most information and images taken from [Learn the Autodesk Fusion User Interface]()

1.1 The interface



You will find it very difficult to navigate this interface without a mouse. ALWAYS come prepared with one, or borrow from a colleague / another room.

1.1.1 **Application Bar (1)**

The application bar contains your open project tabs (similar to Chrome), the file menu (used for importing or exporting projects) and a save button.

1.1.2 **Data Panel (2)**

The Data Panel houses all of your design files. Within the data panel, you can create new projects and folders, to further organize your files.

Fusion 360 automatically saves your projects to the cloud every few minutes, ensuring your work is versioned and recoverable even if your computer crashes.

1.1.3 **Profile and Notification Center (3)**

This area contains profile information, such as name and account status, but also provides job status information and notifications.

While most of our work in Fusion will be handled by our computers, some jobs (most commonly exporting and STL file) will be done in Autodesk’s cloud. A notification will appear in this area updating you on the status of such jobs.

1.1.4 **Toolbar (4)**

The toolbar is the most used section while working in Fusion 360. It dynamically changes based on your active workspace and organizes tools into tabs. As you start to discover your own common workflows you can customize and rearrange your toolbar features.

1.1.5 **Browser (5)**

The browser lists objects in your design, including planes, sketches, parts, assemblies, and so on. You can think of the Browser as your file structure.

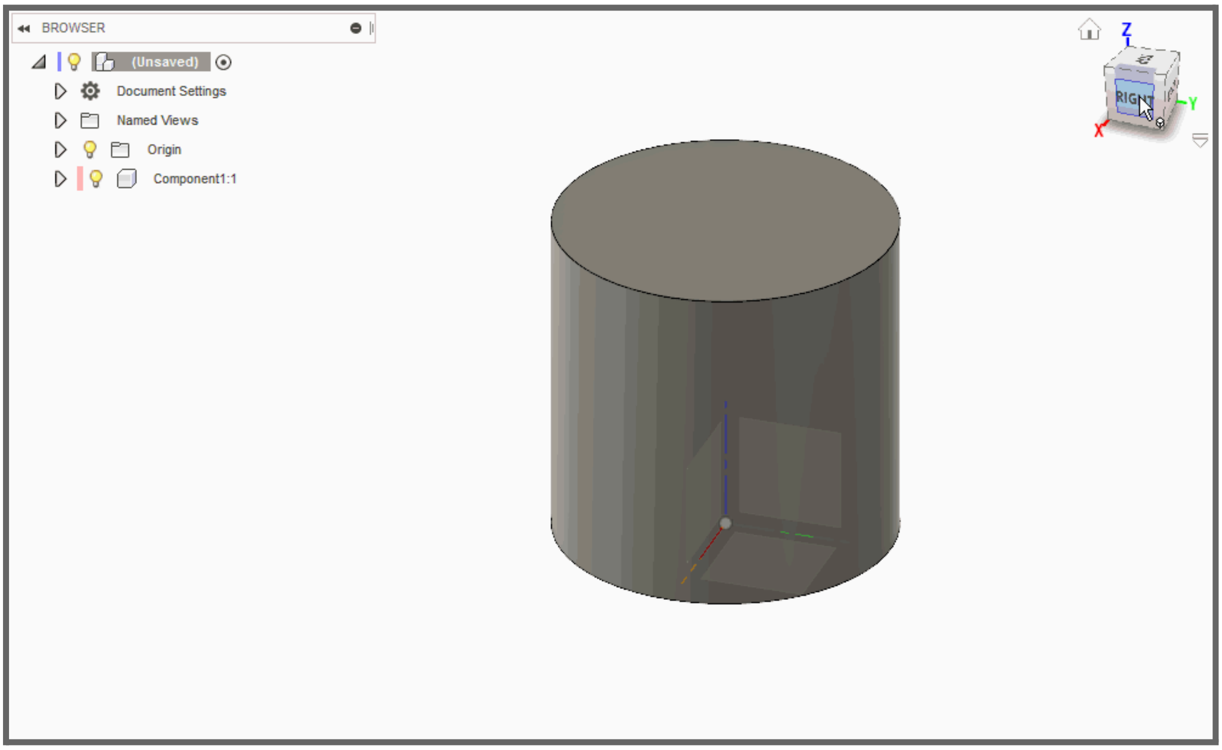
Within the browser, you can change the visibility of objects as well as change your document units. We will also be using it to rename components appropriately and reordering assemblies.

1.1.6 **ViewCube (6)**

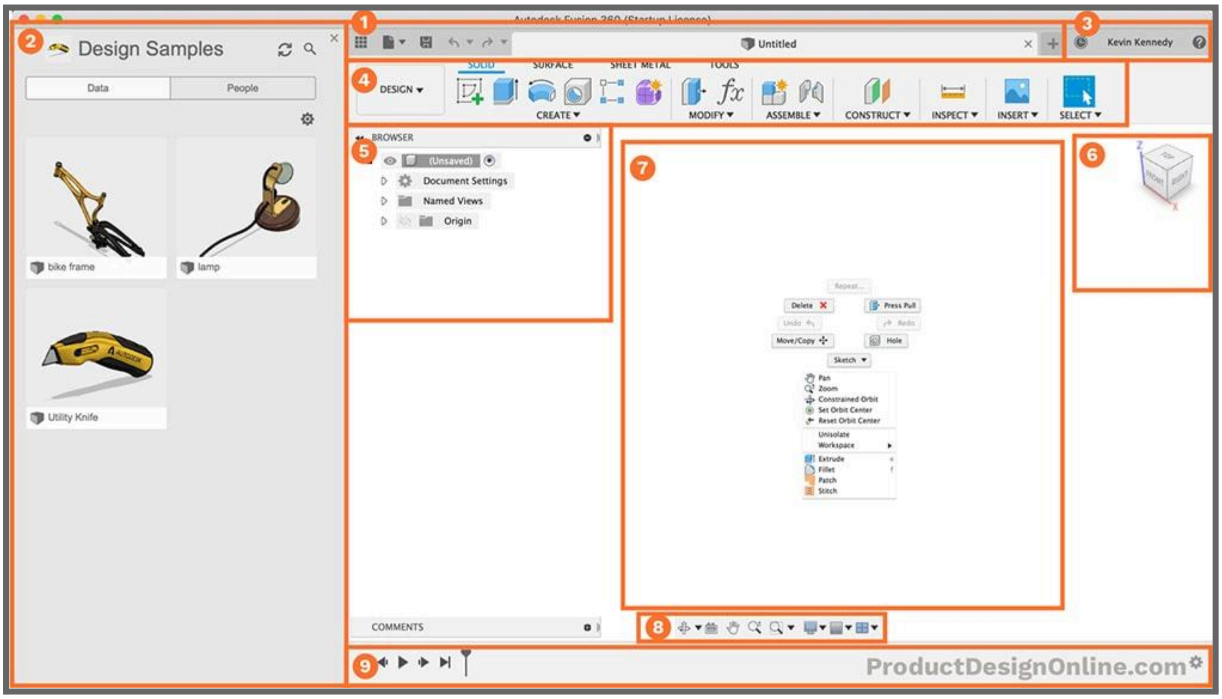
The viewcube allows you to orbit your design or view the design from standard view positions. You can either select faces, corners, or the arrows, or you can simply click and drag the viewcube around.

Exercise: Click and drag on the viewcube to observe the movement. Then click any of the faces / edges. What happens?

> You can also pan around the design by holding the **middle mouse button** and dragging anywhere across the screen. **Shift** + **middle mouse button** rotates your screen just like the viewcube.



You can also readjust the default orientation of the view cube relative to your design.



1.1.7 **Canvas (7)**

The middle section of Fusion 360 is where you'll be sketching and doing all of your design work. Therefore, this section is referred to as the canvas.

1.1.8 **Navigation bar (8)**

The navigation bar contains commands used to zoom, pan, and orbit your design. These options will give you a little bit more control over the use of the viewcube.

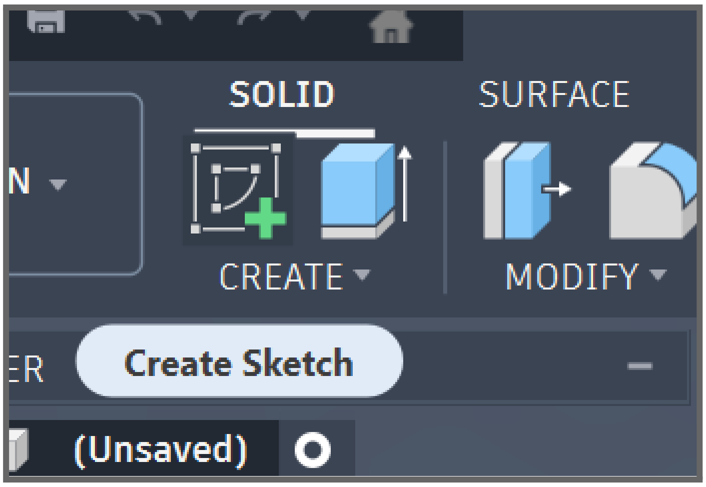
1.1.9 **Timeline (9)**

The timeline lists the order of operations performed on your design. Double-click on timeline features to quickly edit their properties. You can also right-click operations to make additional changes.

Because Fusion 360 is a parametric modeling program, you can also drag the operations around to change the order they are calculated. However, you'll want to be very careful as changing the order can also cause errors or problems with your model.

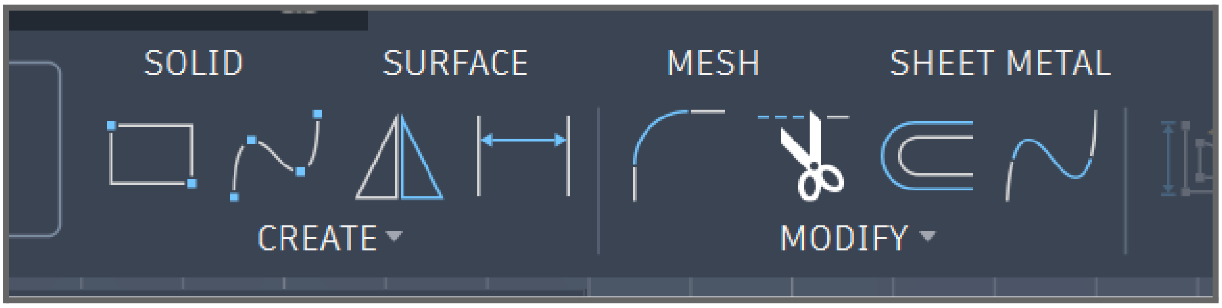
2. Our First Design

2.1 Sketching



In order to start parametric modeling, we will first need to create a sketch. A sketch in Fusion 360 is a 2D geometric profile—drawn with lines, circles, and constraints on a plane—that serves as the foundation for creating 3D features like extrusions.

Make sure you're in the Design workspace (top-left corner). Click Create Sketch in the toolbar—Fusion will automatically highlight the three origin planes (XY, YZ, XZ). Pick any plane you wish to set as the base of your model. The view snaps to that plane, Sketch tab will appear up top, and the Sketch Palette shows on the right.



The toolbar will now contain a different set of tools used for sketching. Here are the ones we will be using today:

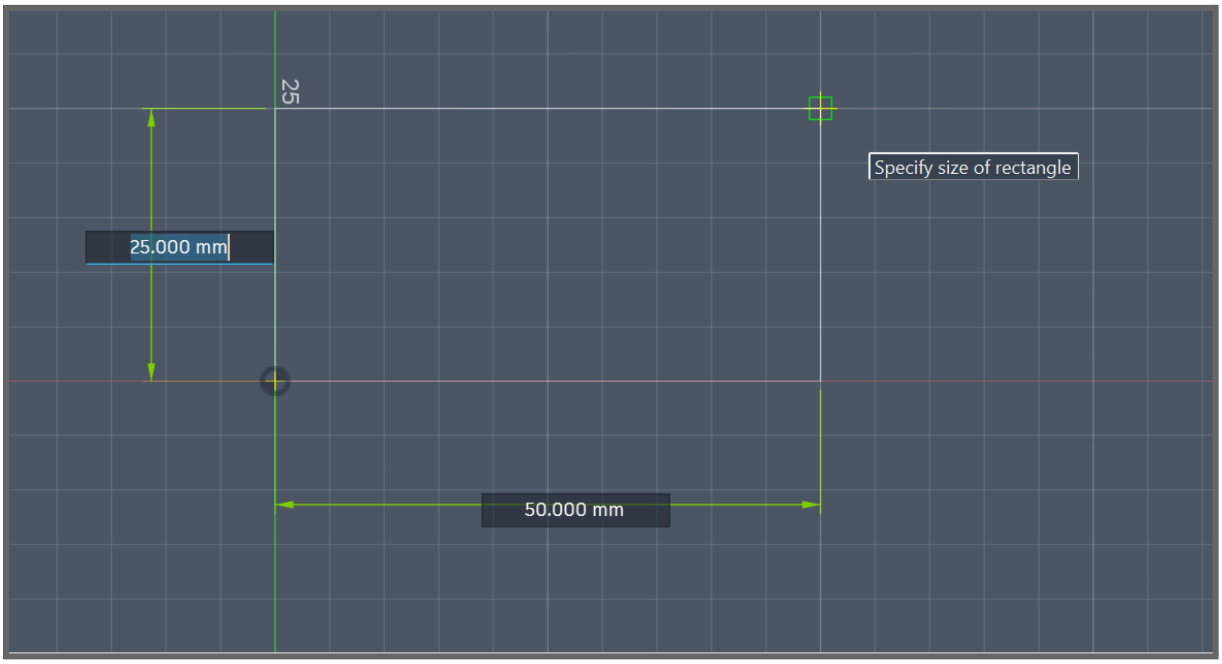
- **Rectangle**
  The Rectangle tool in Fusion 360 creates precise rectangular profiles using methods like 2-Point (corner-to-corner), 3-Point (angled), or Center Rectangle (origin-based)—ideal for boxes, plates, or base shapes in sketches.

- **Line**
  The Line tool draws straight line segments between points, automatically applying constraints like horizontal/vertical when snapped. Connects points to build custom shapes.

- **Circle**
  The Circle tool generates circular with different constraints (center point, tangent or double tangent).

- **Sketch Dimension**
  Sketch Dimension adds editable measurements (lengths, angles, radii) to sketch entities. It turns freehand sketches into precise, parametric designs.

Some of these tools may be hidden under the “CREATE” dropdown.



Selecting any of the first 3 tools will allow you to draw the selected shape pressing **Click + drag** to adjust the size of the shape. While drawing any shape, the “Sketch Palette" will contain relevant settings for the shape being drawn.

While drawing a shape, you will also be able to see the dimensions of the shape being drawn. You can adjust these dimensions by dragging your cursor or manually inputting them.

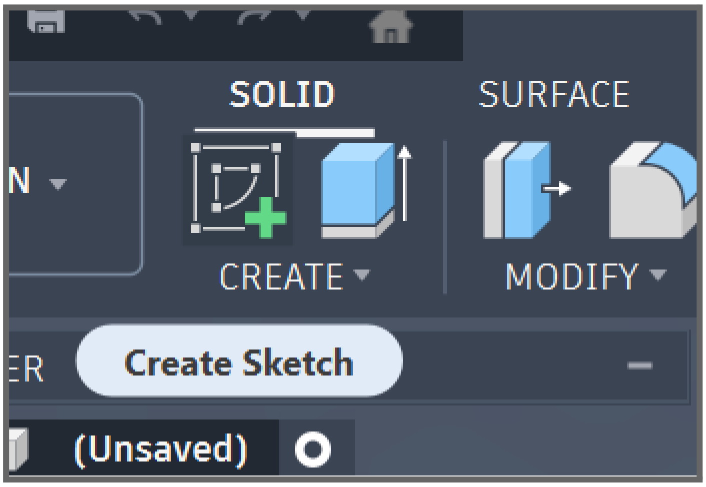
When you're satisfied with the rectangle, go to the top toolbar and click Finish Sketch to go back into the “**Solid**” tab.

2.2 Extruding

In order to turn our 2D rectangle into a 3D object we will now need to “project” it into a rectangular block of a specified thickness. This is called “**Extruding**”.

“**Extrude**” is the second option in our toolbar, right next to “Create Sketch”. Select it and click inside the profile you wish to extrude.

An arrow will appear, allowing you to set the thickness of your object. Please note that the extrude function has multiple options on the right of your screen.



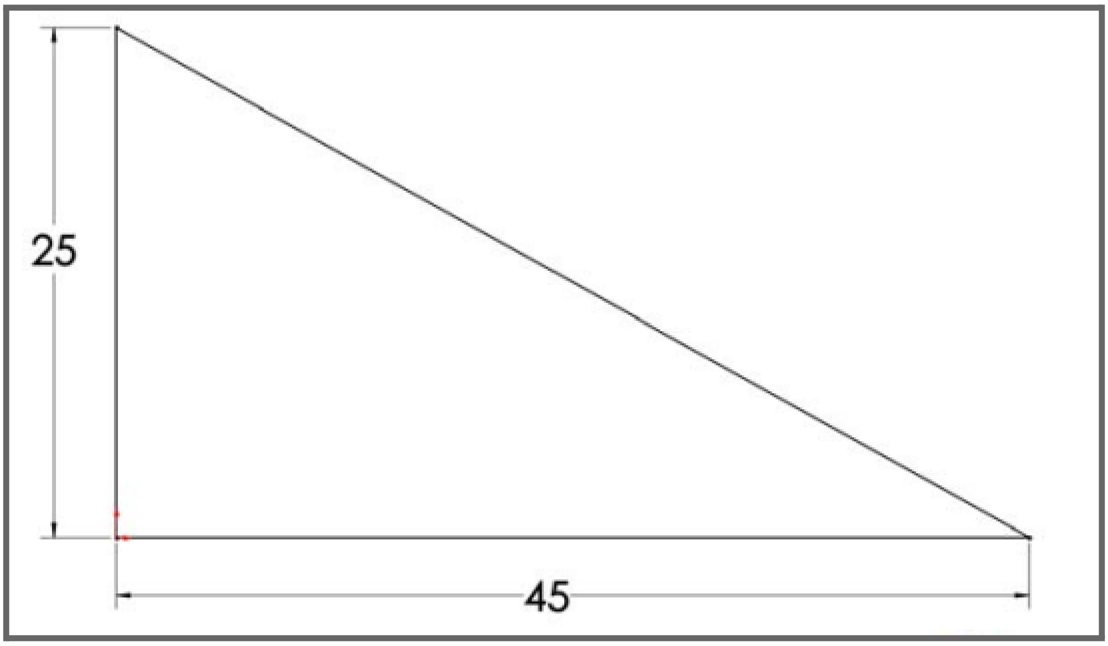
You can access “Extrude” faster by pressing **E** on your keyboard.

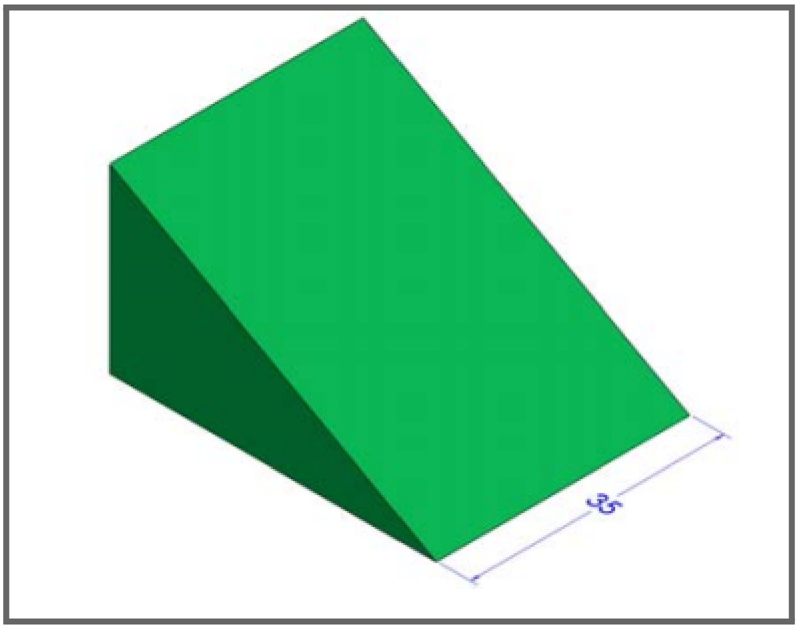
After pressing “Ok” to finish extruding you will now have a 3D prism.

3. Exercises

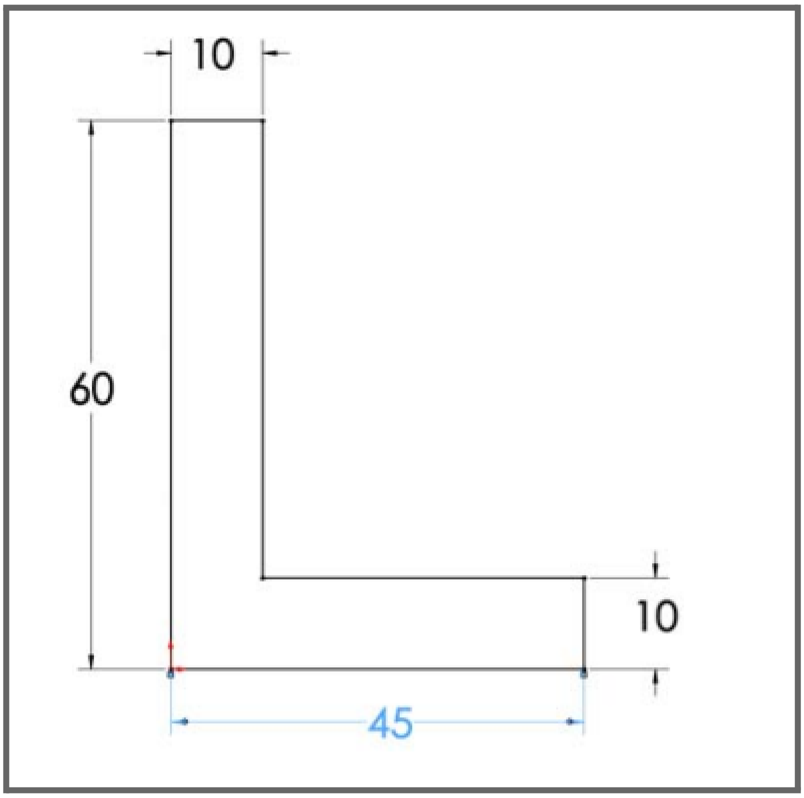
Using sketching and extruding, recreate the following shapes:

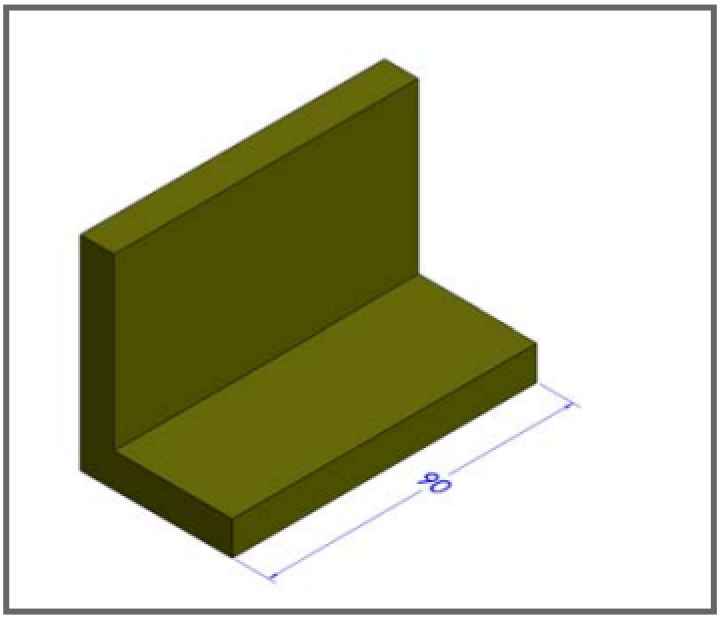
3.1 Exercise 1



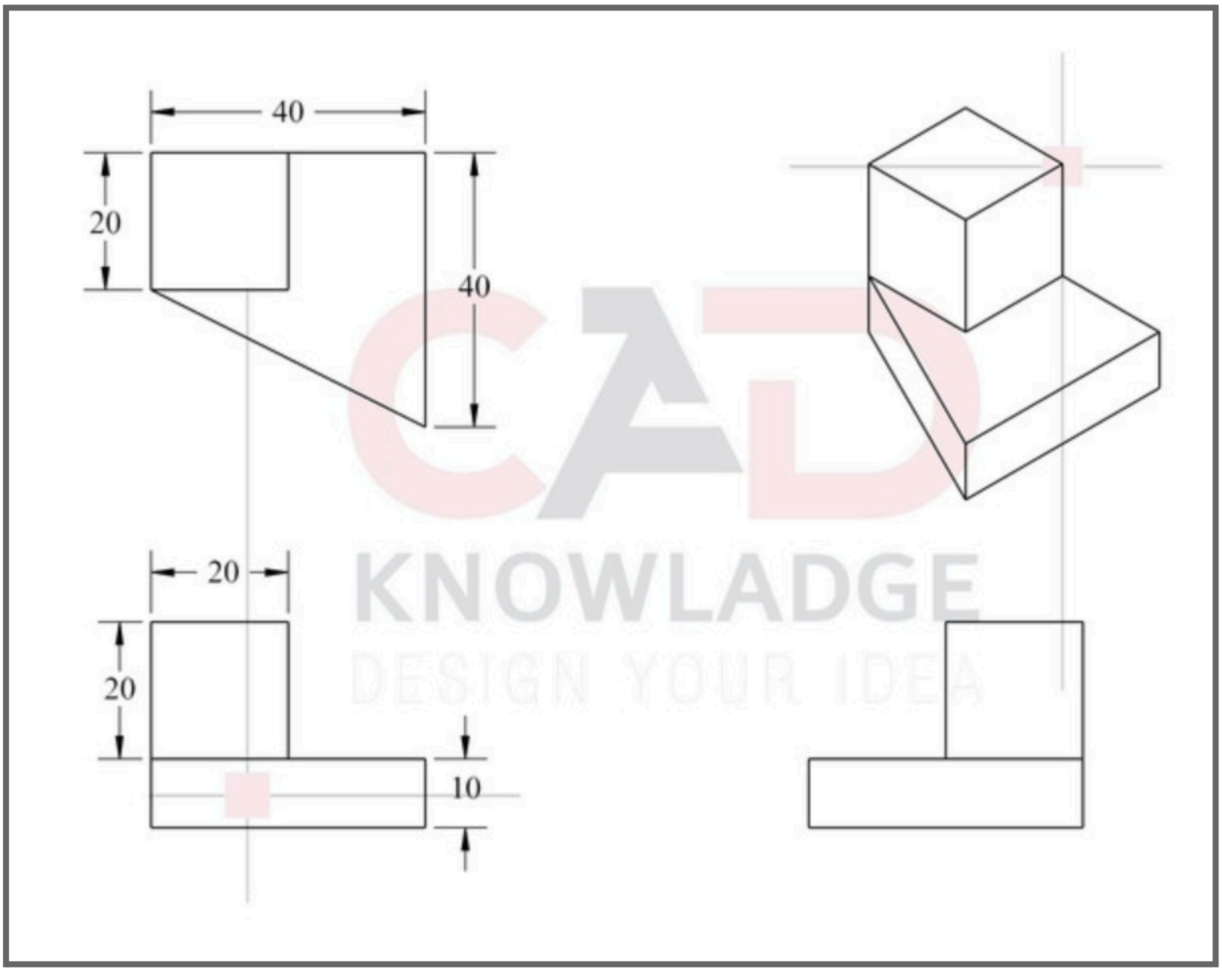


3.2 Exercise 2

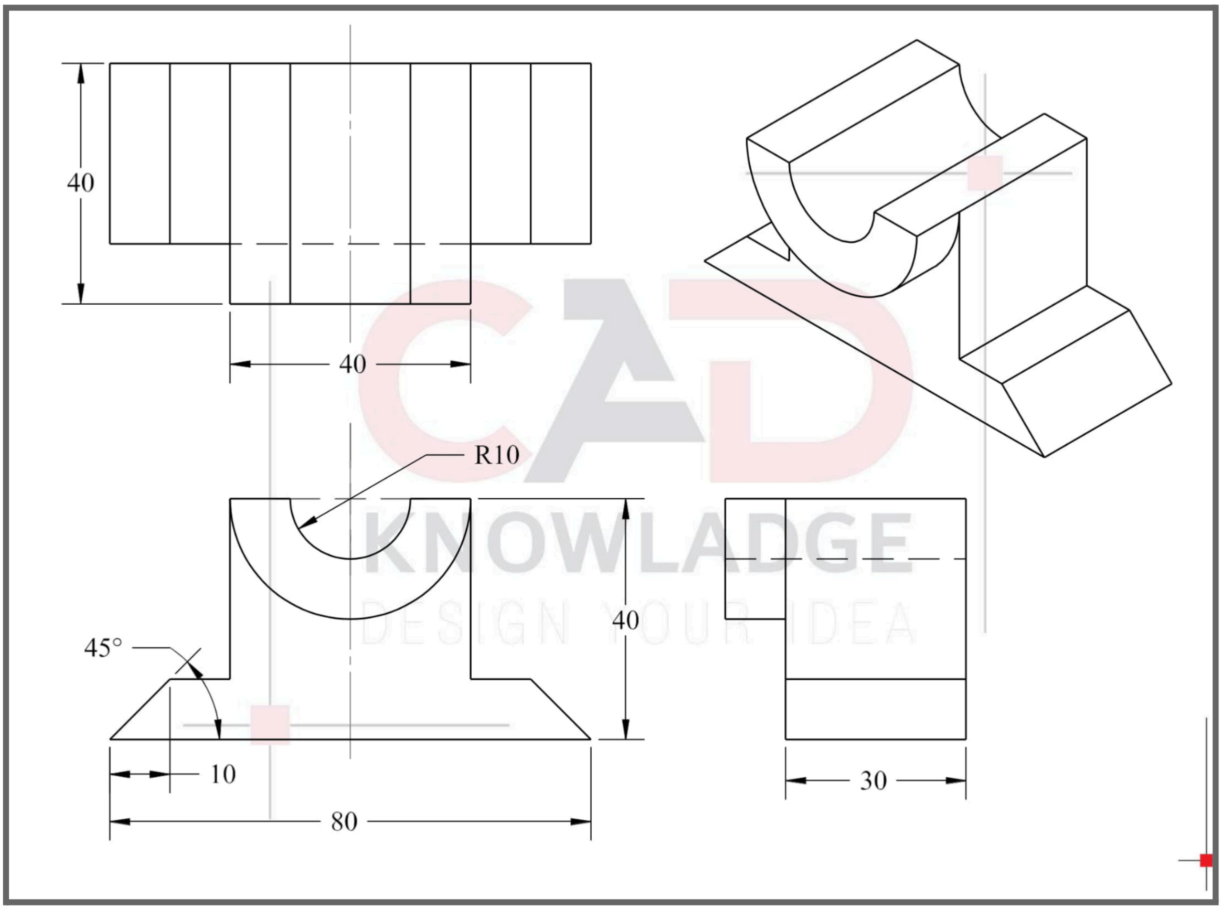




3.3 Exercise 3



3.4 Exercise 4



3.5 Exercise 5